

Practice Exercises and Case Packet

AI Concepts for Official Statisticians

OS-A02-A07 | Validated Candidate Version 1.0 | 26 September 2026

Purpose

This packet develops the ability to classify AI mechanisms, decompose official-statistics tasks, test evidence, diagnose failure propagation and issue bounded institutional decisions. It contains no answers. The exercises progress from baseline diagnosis to independent postgraduate performance.

Control	Learner requirement
Permitted inputs	Public, synthetic, de-identified or formally approved controlled material only.
Evidence rule	Record what was tested, against which reference, for which population and tolerance.
Decision vocabulary	Permit bounded pilot, redesign, reject, pause or escalate.
Non-compensatory duties	Authority, confidentiality, statistical quality, reproducibility and professional independence.
Submission	One traceable dossier plus requested worksheets; do not substitute fluent prose for evidence.

Use with the laboratory

Exercises 9 and 10 use outputs from OS-A02-A06. A laboratory result may be questioned, but any alternative conclusion must cite the exact evidence and tolerance used.

Evidence protocol

Step	Question	Required record
1 Purpose	Which statistical decision or product is affected?	Purpose, user and consequence.
2 Task	What operations and judgements occur?	Task decomposition.
3 Mechanism	What produces the output?	Mechanism family and counterexample.
4 Evidence	What independent test is relevant?	Population, reference, metric and tolerance.
5 Failure	How can error propagate?	Failure chain and affected groups.
6 Control	What prevents, detects or contains failure?	Owner, timing and stopping rule.
7 Decision	What is permitted now?	Bounded decision, conditions and residual uncertainty.

Answer boundary

This learner packet deliberately withholds solutions, thresholds for case-specific judgements and facilitator interpretations. Do not seek a single technology preference. The assessable work is the quality and traceability of your reasoning.

Part I Baseline classification

Complete Exercises 1 to 4 before consulting the canonical lesson or laboratory. Preserve your first responses, then annotate revisions after study.

Exercise 1 Mechanism identification

A survey platform rejects an age of 213; a fitted model predicts nonresponse; a language model drafts a release paragraph; a retrieval system answers from approved metadata; an optimizer allocates interviewer routes; and a tool-using system sends reminders.

Required work

1. Classify each mechanism as rule-based, supervised or statistical learning, generative, retrieval-augmented, optimization, agentic/tool-using, or not enough information.
2. For each classification, identify the observable fact that supports it.
3. Write one additional fact that could change your classification.
4. Separate technical capability from institutional authority.

Evidence to submit

- Six-row classification matrix
- Uncertainty statement for every ambiguous item

Response record

Exercise 2 Automation is not one category

A director describes all six systems in Exercise 1 as AI and asks for one common accuracy target.

Required work

1. Explain why the common label does not establish a common evaluation design.
2. For three mechanisms, define different units of analysis and failure events.
3. Identify which outputs require factual, inferential, procedural or authority validation.
4. Draft a two-sentence correction suitable for senior management.

Evidence to submit

- Evaluation-design comparison
- Management correction

Response record

Exercise 3 Task decomposition

The labour-statistics unit wants to automate occupation coding from free-text job descriptions.

Required work

1. Decompose the workflow from description receipt through code assignment and publication.
2. Mark deterministic operations, empirical inference, ambiguous judgement and release authority.
3. Identify the smallest bounded component that could be tested first.
4. State what the system must never decide without designated official authority.

Evidence to submit

- Workflow map
- Boundary statement
- Proposed testable component

Response record

Exercise 4 Evidence claim audit

A vendor states that its coding assistant is 96 percent accurate and has been used successfully by several institutions.

Required work

1. List the information missing from the 96 percent claim.
2. Define a reference standard and sampling design for independent verification.
3. Specify at least three subgroup analyses and justify them.
4. Identify a material error that overall accuracy could conceal.
5. State the minimum evidence needed before a bounded pilot could be considered.

Evidence to submit

- Claim audit
- Independent evaluation specification

Response record

Part II Guided Kintaba cases

Use the evidence protocol. Each case includes enough information to support competing hypotheses, not a predetermined answer.

Exercise 5 Occupation coding candidate assistant

Kintaba reports 91.2 percent manual coding accuracy. A proposed supervised classifier reports 94.6 percent overall accuracy. Recall is 81.3 percent for informal work and 76.8 percent in the two smallest language strata. The proposed tolerance is 90 percent recall for each priority group.

Required work

1. State the proposed use and exclude at least two broader uses.
2. Assess the aggregate and subgroup evidence against the stated tolerance.
3. Trace how coding errors could affect published distributions and revision workload.
4. Propose controls, a manual route and a stopping rule.
5. Issue and defend a bounded decision.

Evidence to submit

- Completed task and technology classification
- Subgroup evidence table
- Failure chain
- Decision paragraph

Response record

Exercise 6 Probabilistic record linkage

The business register team proposes a model to link survey units with tax records. A validation sample gives 97 percent precision and 88 percent recall. False links are concentrated among enterprises sharing addresses, while missed links are concentrated among small informal units.

Required work

1. Distinguish false links from missed links and their consequences for the intended use.
2. Explain why high precision alone does not validate frame updating.
3. Define an adjudicated reference sample and uncertainty reporting plan.
4. Compare analytical linkage with automatic register updating.
5. Issue separate decisions for research analysis and operational updating.

Evidence to submit

- Loss matrix
- Population and subgroup test plan
- Two bounded decisions

Response record

Exercise 7 Crop area estimation

A satellite classifier improves pixel accuracy from 86 to 91 percent. Its errors are spatially clustered around small mixed-crop plots. The agricultural survey uses calibrated area estimates by district.

Required work

1. Explain why pixel accuracy is not the official estimand.
2. Draw the chain from imagery to classification, area aggregation, calibration and published estimate.
3. Specify reference data, spatial validation and design-based checks.
4. Identify how clustered errors could bias district estimates.
5. Choose between pilot, redesign, rejection or escalation and state conditions.

Evidence to submit

- Estimand statement
- Error-propagation diagram
- Validation design
- Conditional decision

Response record

Exercise 8 Release drafting assistant

A retrieval-assisted language model drafts statistical highlights from approved tables and metadata. In a 50-release trial it reduces median drafting time by 35 percent. Four drafts reverse a comparison, six cite irrelevant metadata and reviewers spend 20 percent more time verifying claims.

Required work

1. Define a quality-adjusted time measure rather than raw drafting time.
2. Classify factual, interpretive, citation and release-authority failures.
3. Design a claim-level verification sample and critical-error rule.
4. Specify corpus governance, logging and reviewer responsibilities.
5. State whether the evidence supports continued experimentation, a bounded pilot or rejection.

Evidence to submit

- Quality-adjusted efficiency calculation
- Claim verification protocol
- Bounded decision

Response record

Part III Laboratory evidence challenge

Open OS-A02-A06 and reproduce the reference run before answering. Record software versions, seed, data files and any deviation.

Exercise 9 Baseline versus learned classifier

The executed laboratory reports that the transparent rule baseline outperforms the learned classifier on the synthetic task-classification problem.

Required work

1. Reproduce the reported baseline and model macro F1 values.
2. Explain why a more complex mechanism is not justified merely because it is learned.
3. Identify at least two reasons the synthetic data favour the rule baseline.
4. Propose one legitimate redesign of the data-generating or evaluation process without manipulating the test result.
5. State what decision is justified now.

Evidence to submit

- Reproduction record
- Model comparison
- Redesign proposal

Response record

Exercise 10 Subgroup and stress failure

The laboratory reports an unseen-region stress test and a regional macro-F1 range that exceeds the stated tolerance.

Required work

1. Locate the precise evidence that causes the gate to fail.
2. Explain why overall performance cannot compensate automatically for the subgroup breach.
3. Distinguish a data-coverage problem, a model problem and a task-definition problem.
4. Design remediation evidence and a re-entry criterion.
5. Write a short briefing to the statistical-methods committee.

Evidence to submit

- Gate audit
- Root-cause hypotheses
- Remediation and re-entry plan
- Committee briefing

Response record

Part IV Challenge variations

For each variation, revise—not merely repeat—your earlier analysis.

Variation	Required response
A model supplier changes the underlying model without notice.	Identify invalidated evidence, close or preserve the gate, and specify equivalence testing.
A questionnaire redesign changes job-description wording.	Assess drift, reference labels and retraining authority.
A cited metadata page is superseded.	Withdraw affected claims and repair corpus governance.
Reviewer workload doubles during release week.	Recalculate feasible controls and quality-adjusted efficiency.
A minority-language recall falls below tolerance.	Restrict use, preserve access and define remediation evidence.
A prompt injection triggers an unauthorized tool call.	Contain, preserve evidence, revoke permissions and test recovery.
Management requests publication despite a failed gate.	State the statistician’s duty, escalation route and documentation.

Exercise 11 Variation memorandum

Choose any three variations. For each, submit a one-page change memorandum covering evidence invalidated, immediate action, responsible authority, new test and decision status.

Part V Independent cases

Complete one case without procedural prompting. State reasonable assumptions, but never invent evidence.

Exercise 12 Public statistical query service

An agency proposes a public chatbot using approved publications, metadata and methodological manuals. Early tests show fluent answers, 89 percent correct citations and inconsistent refusals when no current statistic exists.

Required work

1. Define intended users, supported questions and excluded decisions.
2. Design a reference question set with rare and adversarial cases.
3. Specify entailment, currency, completeness, refusal and accessibility measures.
4. Define mandatory referral and correction procedures.
5. Issue a conditional institutional decision and monitoring plan.

Evidence to submit

- Full classification dossier
- Evaluation matrix
- Incident and correction workflow

Response record

Exercise 13 Fieldwork support agent

A tool-using assistant can read assignment files, propose interviewer routes, send reminders and escalate nonresponse. It cannot access response content, but it can change schedules and contact timing.

Required work

1. Map every action, permission and affected party.
2. Separate optimization, rule execution and agentic orchestration.
3. Identify legal, operational, burden and exclusion risks.
4. Specify least privilege, approval points, logs, rollback and incident controls.
5. Decide which actions, if any, may be automated.

Evidence to submit

- Permission matrix
- Failure and rights analysis
- Action-by-action decision

Response record

Exercise 14 Administrative data anomaly triage

A ministry supplies monthly benefit records. An anomaly detector flags unusual changes for investigation. Historical labels are incomplete because earlier investigations focused on large regions.

Required work

1. Define whether the target is error, change, fraud, reporting delay or investigative priority.
2. Explain the selection bias in historical labels.
3. Construct a validation design that includes small regions and uninvestigated cases.
4. Specify how flags affect editing, imputation and published estimates.
5. Issue a decision that preserves professional independence.

Evidence to submit

- Target-definition note
- Sampling and reference plan
- Propagation analysis
- Decision

Response record

Part VI Reusable evidence worksheets

Worksheet A Task and technology classification

Field	Learner record
Statistical purpose and user	
Unit, population and time	
Workflow boundary	
Inputs and provenance	
Output and intended use	
Mechanism family	
Learning or adaptation	
Decision authority	
Explicit exclusions	
Uncertainty	

Worksheet B Evaluation design

Element	Specification
Reference standard	
Sampling or challenge-set design	
Baseline comparator	
Metric and denominator	
Tolerance and rationale	
Subgroups and rare cases	
Stress and recovery test	
Independent verifier	

Worksheet C Failure propagation register

Failure event	Immediate effect	Downstream statistic	Affected group	Prevention	Detection	Containment	Owner

Worksheet D Bounded decision record

Decision field	Learner record
Decision: permit, redesign, reject, pause or escalate	
Authorized scope	
Conditions before use	
Non-compensatory gate	
Monitoring indicator and tolerance	
Stopping and rollback rule	
Residual uncertainty	
Named authority and review date	

Submission and self-audit

Check	Yes or revision required
I classified the mechanism from observable behaviour, not its product label.	
I separated component performance from fitness for official use.	
I reported denominators, populations, subgroups and uncertainty.	
I used an independent reference and a credible baseline.	
I traced at least one error into a statistical product or decision.	
I assigned controls and stopping authority to identifiable functions.	
I did not use confidential or identifiable data without authorization.	
I did not treat human review as sufficient without time, evidence and power.	
My decision is bounded and could change when specified evidence changes.	
I documented what remains unknown.	

Oral defence questions

- Which evidence had the greatest influence on your decision?
- Which assumption is most fragile?
- What result would reverse your decision?
- Which condition cannot be traded for productivity?
- Why is your chosen baseline credible?
- Who bears the residual risk and who has authority to stop?

Controlled source basis

- SRAI OS-A02-A01 Canonical Lesson v0.2.
- SRAI OS-A02-A03 Learner Guide v1.0.
- SRAI OS-A02-A04 Facilitator Guide v1.0.
- SRAI OS-A02-A05 Presentation and Narration v1.0.
- SRAI OS-A02-A06 Executable Classification Laboratory v1.0.

Solutions and assessor guidance are intentionally excluded and belong to OS-A02-A08.